

Department of Computer Engineering
Academic Term: July-November 2022

Class : B.E. Computer
Semester : VII

Subject Name :ML
Subject Code : CSC701

Practical No:	7
Title.	To Study and implement Ensemble Techniques
Date of Performance.	19/09/2025
Roll No:	9913
Name of the Student:	Mark Lopes

Evaluation

Indicator	Very Poor	Poor	Average	Good	Excellent
Timeline (2)	More than three sessions late (0)	More than two sessions late (0.5)	Two sessions late (1)	One session late (1.5)	Early or ontime (2)
Efforts(4)	N/A	N/A	Not Completed (1)	Partially Completed (2)	Completed(4)
Legibility(2)	N/A	N/A	Poor(1)	Good(1.5)	Very Good(2)
Oral Assessment (2)	N/A	N/A	N/A	Partially Understood the concept (1)	Understood the concept(2)

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Total Marks

Signature of the Teacher

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0

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Experiment 7

To Study and implement Ensemble Techniques

II-THEORY

Introduction To Ensemble Learning

Ensemble Learning Techniques in Machine Learning, Machine learning models suffer bias and/or variance. Bias is the difference between the predicted value and actual value by the model. Bias is introduced when the model doesn't consider the variation of data and creates a simple model. The simple model doesn't follow the patterns of data, and hence the model gives errors in predicting training as well as testing data i.e. the model with high bias and high variance.

When the model follows even random quirks of data, as pattern of data, then the model might do very well on training dataset i.e. it gives low bias, but it fails on test data and gives high variance.

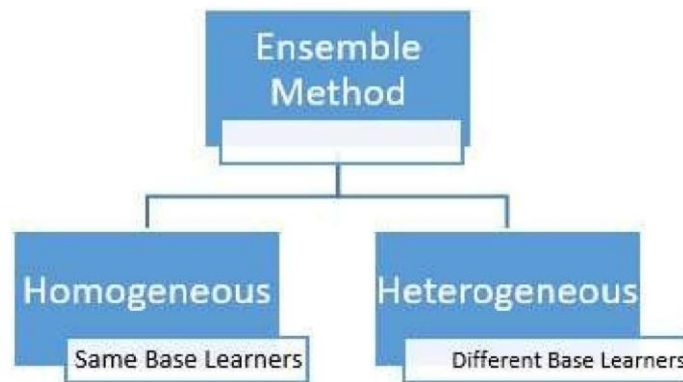
Therefore, to improve the accuracy (estimate) of the model, ensemble learning methods are developed. Ensemble is a machine learning concept, in which several models are trained using machine learning algorithms. It combines low performing classifiers (also called as weak learners or base learner) and combine individual model prediction for the final prediction.

On the basis of type of base learners, ensemble methods can be categorized as homogeneous and heterogeneous ensemble methods. If base learners are same, then it is a homogeneous ensemble method. If base learners are different then it is a heterogeneous ensemble method.

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Ensemble Learning Methods

Ensemble techniques are classified into three types:

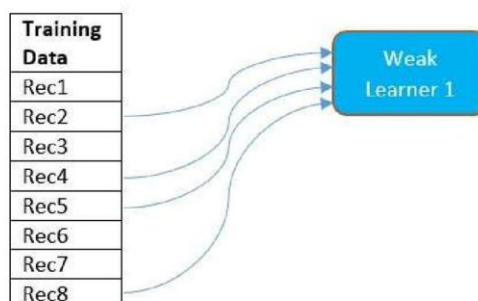
1. Bagging
2. Boosting
3. Stacking

Bagging

Consider a scenario where you are looking at the users' ratings for a product. Instead of approving one user's good/bad rating, we consider average rating given to the product. With average rating, we can be considerably sure of quality of the product. Bagging makes use of this principle. Instead of depending on one model, it runs the data through multiple models in parallel, and average them out as model's final output.

What is Bagging? How it works?

- Bagging is an acronym for **Bootstrapped Aggregation**. Bootstrapping means random selection of records with replacement from the training dataset. 'Random selection with replacement' can be



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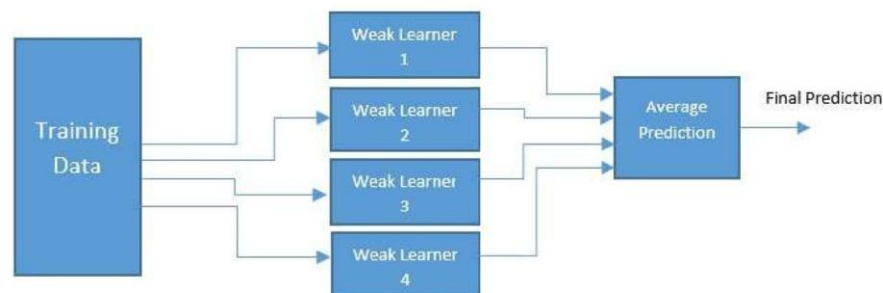
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explained as follows:

- Consider that there are 8 samples in the training dataset. Out of these 8 samples, every weak learner gets 5 samples as training data for the model. These 5 samples need not be unique, or non-repetitive.
 - The model (weak learner) is allowed to get a sample multiple times. For example, as shown in the figure, Rec5 is selected 2 times by the model. Therefore, weak learner1 gets Rec2, Rec5, Rec8, Rec5, Rec4 as training data.
 - All the samples are available for selection to next weak learners. Thus all 8 samples will be available for next weak learner and any sample can be selected multiple times by next weak learners.
- Bagging is a **parallel method**, which means several weak learners learn the data pattern independently and simultaneously. This can be best shown in the below diagram:



- The output of each weak learner is averaged to generate final output of the model.
- Since the weak learner's outputs are averaged, this mechanism helps to reduce variance or variability in the predictions. However, it does not help to reduce bias of the model.
- Since final prediction is an average of output of each weak learner, it means that each weak learner has equal say or weight in the final output.

To summarize:

- Bagging is Bootstrapped Aggregation
- It is Parallel method
- Final output is calculated by averaging the outputs produced by individual weak learner
- Each weak learner has equal say
- Bagging reduces variance

Boosting

We saw that in bagging every model is given equal preference, but if one model predicts data more correctly than the other, then higher weightage should be given to this model over the other. Also, the model should attempt to reduce bias. These concepts are applied in the second ensemble method that we are going to learn, that is Boosting.

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What is Boosting?

1. To start with, boosting assigns equal weights to all data points as all points are equally important in the beginning. For example, if a training dataset has N samples, it assigns weight = $1/N$ to each sample.
2. The weak learner classifies the data. The weak classifier classifies some samples correctly, while making mistake in classifying others.
3. After classification, sample weights are changed. Weight of correctly classified sample is reduced, and weight of incorrectly classified sample is increased. Then the next weak classifier is run.
4. This process continues until model as a whole gives strong predictions.

Note: Adaboost is the ensemble learning method used in binary classification only.

How Adaboost Algorithm Works?

Consider below training data for heart disease classification.

Blocked Arteries	Chest Pain	Weight	Heart Disease
Y	Y	200	Y
Y	N	185	Y
N	Y	200	Y
Y	Y	160	Y
Y	N	140	N
Y	N	130	N
N	Y	170	N
Y	Y	170	N

1. Initialize Weights To All Training Points

First step is to assign equal weights to all samples as all samples are equally important. Always, sum of weights of all samples equals 1. There are 8 samples, so each sample will get weight = $1/8 = 0.125$

Since all samples are equally important to start with, all samples get equal weight: $1 / \text{total number of samples} = 1/8$

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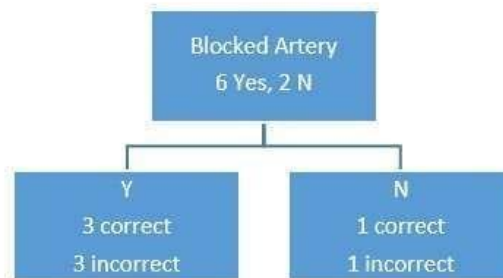
Blocked Arteries	Chest Pain	Weight	Heart Disease	Weights Assigned
Y	Y	200	Y	1/8
Y	N	185	Y	1/8
N	Y	200	Y	1/8
Y	Y	160	Y	1/8
Y	N	140	N	1/8
Y	N	130	N	1/8
N	Y	170	N	1/8
Y	Y	170	N	1/8

2. Create Stump

After assigning weight, next step is to create stumps. Stump is a decision tree with one node and two leaves. Adaboost creates forest of decision stumps. To create stump, only one attribute should be chosen. But, it is not randomly selected. The attribute that does the best job of classifying the sample is selected first.

So let's see how each attribute classifies the samples.

Blocked Artery:

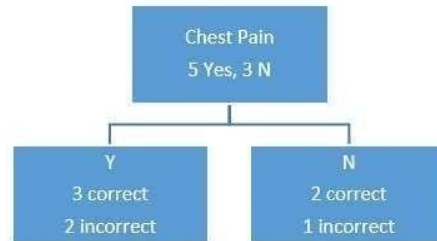


Chest Pain

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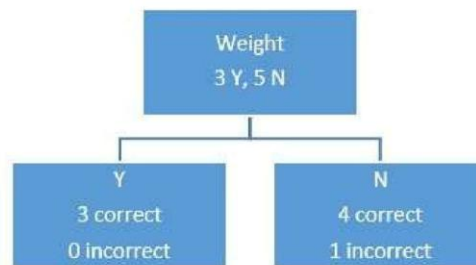
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Weight:

Weight is continuous quantity, and there is a method to find threshold for continuous quantity in decision tree algorithm. According to that method, the threshold is 175. Based on this condition draw the decision stump.



From above classification, it can be seen that

- 'Blocked Arteries' as a stump made 4 errors.
- 'Chest Pain' as a stump made 3 errors.
- 'Weight' as a stump made 1 error.

Since, the attribute 'Weight' made least number of errors, 'Weight' is the best stump to start with.

Note: Since this was attribute selection for first stump, we need not consider weight of each sample. This is because, to start with, all samples have equal weights.

To select next stump, both, weight of samples and errors made by that attribute while classifying the samples, will be considered.

3. Calculate Total Error And Voting Power (Amount Of Say) Of The Stump

After choosing best attribute for stump, next step is to calculate the voting power (amount of say) of that stump in the final classification. Formula to calculate amount of say is:

$$\text{Amount of say} = \frac{1}{2} * \ln\left(\frac{1 - \text{Total Error}}{\text{Total Error}}\right)$$

Where $\ln$ is Natural Log

Total Error = Number of misclassified samples * Weight of sample.

Since the stump misclassified only 1 sample, Total Error made by stump = weight of 1 sample = $1/8 = 0.125$

Substituting Total Error in the equation,

Amount of say = $\frac{1}{2} * \ln((1 - 0.125)/0.125)$

= 0.9729

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4. Start To Build Final Classifier Function

The Adaboost classifier then starts building a function that will assign this 'amount of say' as voting power of the stump. It will assign 0.9729 as voting power to the stump created for 'Weight' attribute.

The function $h(x) = 0.9729 * \text{'Weight'}$ as stump

5. Update New Weights For The Classifier

After calculation of amount of say, next step is to create another stump. However, before that adjust the weights assigned to each sample. The samples that were correctly classified in previous stump should get less importance as their classification is over, and the samples that were not correctly classified should get more weight in the next step.

The formula to assign new weight for **correctly classified** samples is:

$$W_{\text{new}} = W_{\text{old}} / 2(1 - \text{Total Weight})$$

The formula to assign new weight for **incorrectly classified** samples is:

$$W_{\text{new}} = W_{\text{old}} / (2 * \text{Total Weight})$$

New weights of correctly classified samples =

$$W_{\text{new}} = (0.128) / 2(1 - 0.125) = 0.07$$

New weights of incorrectly classified samples =

$$W_{\text{new}} = (0.125) / (2 * 0.125) = 0.5$$

Substitute new weights:

Blocked Arteries	Chest Pain	Weight	Heart Disease	Weights Assigned
Y	Y	200	Y	0.07
Y	N	185	Y	0.07
N	Y	200	Y	0.07
Y	Y	160	Y	0.5
Y	N	140	N	0.07
Y	N	130	N	0.07
N	Y	170	N	0.07
Y	Y	170	N	0.07

From above classification, it can be seen that

- 'Blocked Arteries' as a stump made 4 errors. Therefore, Total Error for 'Blocked Arteries' = $0.07 * 4 = 0.28$

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- 'Chest Pain' as a stump made 3 errors. Therefore, Total Error for 'Chest Pain' = $0.07 * 3 = 0.21$
- 'Weight' as a stump made 1 error. Therefore, Total error for weight = $1 * 0.5 = 0.5$.

Note: We have already considered 'Weight' attribute for first stump creation. So this attribute need not be considered again. However, it is considered here to show effect of 'Assigned Weight' of the sample on Total Error calculation.

6. Calculate Voting Power (Amount of Say) and Add to Final Classifier Function

Since Chest Pain classifier has least Total Error, this attribute is selected as second stump.

The Amount of Say = $\frac{1}{2} \ln(1-0.21)/0.21 = 0.66$

Add this classifier with its weight to the function.

The function $h(x) = 0.9729 * \text{'Weight' as stump} + 0.66 * \text{'Chest Pain' as stump}$

7. Assign New Weights and Repeat

Again, assign new weights to the correctly classified samples with formula:

$$W_{\text{new}} = W_{\text{old}} / 2(1 - \text{total weight})$$

And incorrectly classified samples with the formula:

$$W_{\text{new}} = W_{\text{old}} / (2 * \text{total weight})$$

We repeat this process until

1. Enough rounds of this process are done to create strong classifier
2. No good classifiers left for making predictions.

When it comes to classifying data point, even if one classifier misclassifies a point, the voting powers of other classifiers override it making strong predictions.

There is another method to train Adaboost algorithm. In this method, new sample dataset is created in every iteration. The sample dataset is of same size as original dataset, and it contains some samples that are repeatedly selected. Samples with higher weights (i.e. samples that are incorrectly classified in previous round), will be selected repeatedly. The weight vector is normalized.

To summarize:

1. Adaboost is a sequential method where one weak learner runs after other.
2. Weak learners are stumps i.e. decision tree with 2 nodes.
3. Voting power is assigned to stumps based on how correctly it classifies the data.
4. Boosting reduces bias

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Other Boosting Algorithms

Apart from Adaboost, there is other boosting algorithm – Gradient Boosting or Gradient Boosting Machine (GBM). The main idea behind this algorithm is to create decision trees with smaller leaves and scale the tree with a learning rate.

Decision tree algorithm is prone to overfitting and leads to high variance. Hence prediction made by the tree is scaled (or multiplied) with the learning rate. This reduces training accuracy, but provides better results in long run. Gradient Boosting is used in classification as well as regression problems.

The open source implementations Gradient Boosting algorithms are XGBoost, LightGBM that are more regularized, and efficient.

III IMPLEMENT THE FOLLOWING PROBLEM STATEMENTS

IV CODE WITH OUTPUT

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import BaggingClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, classification_report
```

```
df = pd.read_csv("dataset.csv") # ensure dataset.csv is uploaded
df.head()
```

	Unnamed: 0	track_id	artists	album_name	track_name	popularity	duration_ms	explicit	danceability	energy
0	0	5SuOikwiRyPMVolQDJUGSV	Gen Hoshino	Comedy	Comedy	73	230666	False	0.676	0.46
1	1	4qPNDBW1i3p13qLCt0Ki3A	Ben Woodward	Ghost (Acoustic)	Ghost - Acoustic	55	149610	False	0.420	0.16
2	2	1iJBSr7s7jYXzM8EGcbK5b	Ingrid Michaelson;ZAYN	To Begin Again	To Begin Again	57	210826	False	0.438	0.35
3	3	6lfxq3CG4xtTiEg7opyCyx	Kina Grannis	Crazy Rich Asians (Original Motion Picture Soundtrack)	Can't Help Falling In Love	71	201933	False	0.266	0.05
4	4	5vjLSffimilP26QG5WcN2K	Chord Overstreet	Hold On	Hold On	82	198853	False	0.618	0.44

5 rows × 21 columns

```
df = df.drop(columns=["Unnamed: 0", "track_id", "artists", "album_name", "track_name"])

# Encode target variable (track_genre)
label_encoder = LabelEncoder()
df["track_genre"] = label_encoder.fit_transform(df["track_genre"])

# Features (X) and Target (y)
X = df.drop(columns=["track_genre"])
y = df["track_genre"]

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

print("Training set shape:", X_train.shape)
print("Test set shape:", X_test.shape)
```

Training set shape: (91200, 15)
Test set shape: (22800, 15)

```
# =====
# 🚀 Cell 4: Train Bagging Classifier (Fixed)
# =====
bagging_model = BaggingClassifier(
    estimator=DecisionTreeClassifier(), # updated param
    n_estimators=50,                   # number of trees
    random_state=42,
    n_jobs=-1
)

# Train the model
bagging_model.fit(X_train, y_train)

print("✅ Model training complete!")
```

✅ Model training complete!

```
# =====
# 🚀 Cell 5: Evaluate Model (Fixed)
# =====
# Predictions
y_pred = bagging_model.predict(X_test)

# Accuracy
print("Accuracy:", accuracy_score(y_test, y_pred))

# Classification Report (for ALL classes)
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred, target_names=label_encoder.classes_))
```


trance	0.40	0.40	0.40	200
trip-hop	0.29	0.26	0.28	200
turkish	0.25	0.33	0.29	200
world-music	0.37	0.34	0.35	200
accuracy			0.30	22800
macro avg	0.30	0.30	0.30	22800
weighted avg	0.30	0.30	0.30	22800

```
# =====
# 🌟 Cell 6: Predict on New Data (Optional)
# =====
# Example: Take first 5 test samples and predict their genres
sample = X_test.iloc[:5]
predictions = bagging_model.predict(sample)

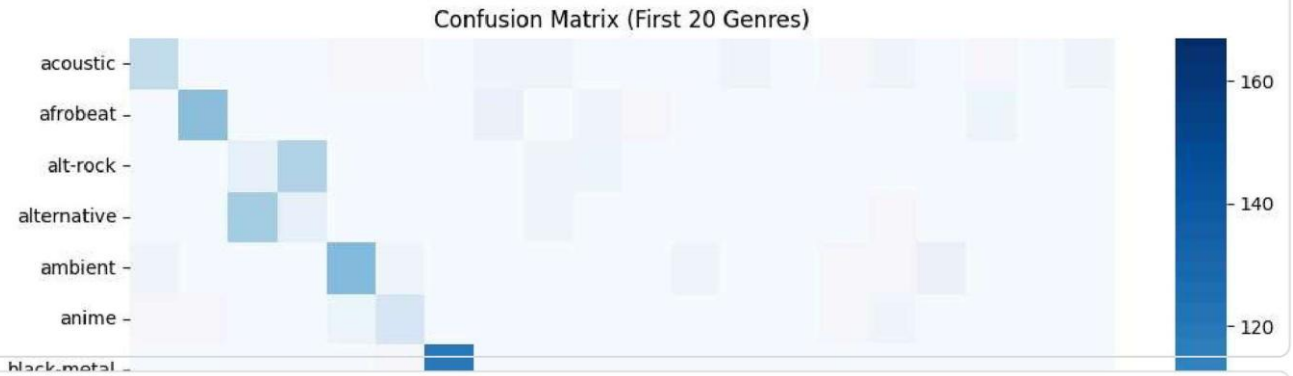
print("Predicted Genres:", label_encoder.inverse_transform(predictions))
```

Predicted Genres: ['disco' 'gospel' 'dancehall' 'j-rock' 'techno']

```
# =====
# 🌟 Cell 7: Confusion Matrix Heatmap
# =====
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import confusion_matrix

# Compute confusion matrix (for all classes)
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(12, 10))
sns.heatmap(cm[:20, :20], cmap="Blues", annot=False, fmt="d",
            xticklabels=label_encoder.classes_[:20],
            yticklabels=label_encoder.classes_[:20])
plt.title("Confusion Matrix (First 20 Genres)")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```

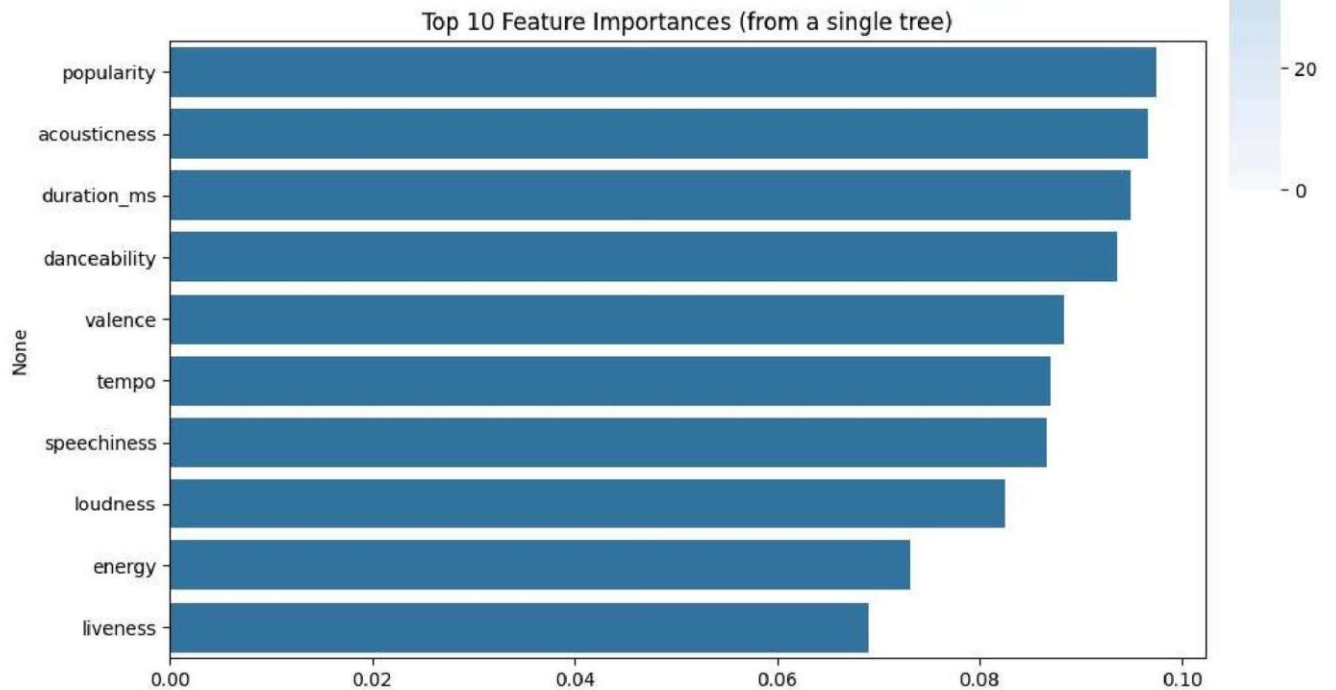


```
# =====
# 🌟 Cell 8: Feature Importance
# =====
import numpy as np

# Since BaggingClassifier doesn't have feature_importances_ directly,
# we can take it from one of its Decision Trees
tree = bagging_model.estimators_[0]

importances = tree.feature_importances_
indices = np.argsort(importances)[::-1]
features = X.columns

plt.figure(figsize=(10,6))
sns.barplot(x=importances[indices][:10], y=features[indices][:10])
plt.title("Top 10 Feature Importances (from a single tree)")
plt.show()
```



Start coding or [generate](#) with AI.

Q.1 what are weak learners?

→ weak learners are simple models in ML that perform just better than random chance. They are usually high bias, low complexity, models meaning they don't capture all patterns in data and individually make many errors.

Q.2 How is random forest related to decision tree?

→ Random forest is directly related to decision tree

i) decision tree: Single tree that makes prediction by splitting data as liked on feedback

ii) Random forest: An ensemble of decision trees. Each tree is trained on random forest subset of data and a random subset of features.

A random forest is like a team of decision trees working together.

Q.3 What is difference between bagging and boosting?

Feature	Bagging	Boosting
i) Goal	Reduce variance	Reduce bias, variance
ii) Method	Train model in parallel	Train model in sequential
iii) Example	Random forest	AdaBoost, GB
iv) Voting	Majoring vote	weight vote